Temporal & Multimodal Visual Intelligence

1. Summary

I treated the task as a controlled research investigation rather than an accuracy competition. The pipeline starts with sequence inspection, moves to pretrained visual representations, and then compares a small temporal GRU against a non-temporal mean-pooled baseline.

The central result is deliberately cautious. The GRU reaches 100% accuracy on the final source-video test split, while the MLP reaches 98.36%. More importantly, shuffling the order of the 16 embeddings leaves the GRU at 100% across ten random seeds. Therefore, the experiment does not demonstrate that the GRU is using temporal order. It is much more likely exploiting visual information already present in individual observations, potentially including source-video or appearance shortcuts.

A progressive-observation test shows that additional observations can help: F1 rises from 0.812 with one observation to 0.988 with two and 1.000 with four. This indicates that more evidence can help, but not necessarily because the model understands temporal relationships. A sequence-level VLM experiment using eight chronological frames also remained confused on aggressive clips.

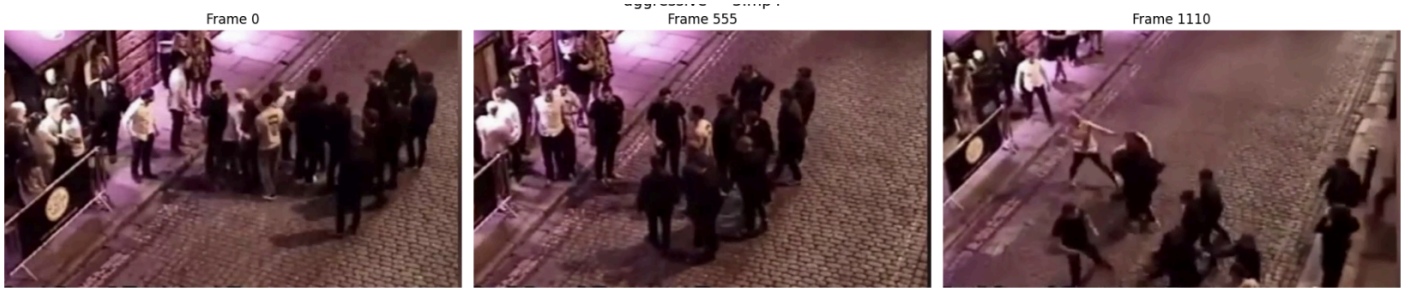
2. Dataset -(Violence/non-violence classification) link :

The dataset contains 11 short videos: five aggressive and six non-aggressive. The source videos differ substantially in frame rate and resolution. Aggressive clips are around 60 FPS at 1920×1080; non-aggressive clips are around 25–30 FPS and generally lower resolution. This is an important confound.

file	class	fps	frame_count	duration_sec	resolution
0.mp4	aggressive	60.032406	741	12.343333	1920x1080
1.mp4	aggressive	60.240964	365	6.059000	1920x1080
2.mp4	aggressive	60.660445	169	2.786000	1920x1080
3.mp4	aggressive	60.063794	1111	18.497000	1920x1080
4.mp4	aggressive	60.058799	1287	21.429000	1920x1080
0.mp4	non_aggressive	29.970030	96	3.203200	1280x720
1.mp4	non_aggressive	30.000000	330	11.000000	960x720
2.mp4	non_aggressive	25.000000	78	3.120000	1280x720
3.mp4	non_aggressive	29.970030	54	1.801800	1174x660
4.mp4	non_aggressive	30.000000	250	8.333333	1280x720

Sequential decoding was verified: videos opened successfully, frames decoded in order, no unexpected read failures were observed, and decoded counts matched metadata. Frame strips showed that some aggressive clips contain frames that are already strongly informative, while other frames are ambiguous. Non-aggressive clips can also contain substantial motion.Temporal &

3. Visualization




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Class: aggressive
Video: 4.mp4
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Class: non_aggressive
Video: 3.mp4
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Observations:
* For aggresive videos There are frames which does not provide necessary information
* In some cases for aggresive videos , we can evidently say by looking at one frame that video is aggresive
* For non-aggresive cases all the frames look similar , no extra ordinary change

Conclusion
* Based on above observation after feature extractions , embeddings change will be maximum for aggresive videos while it should be minimal for non-aggresive cases , This is just an hypothesis , Lets see what our feature extractors tell us

3. PIxel change Analysis

The initial inspection used 16 uniformly sampled frames per video. Raw pixel change was evaluated with temporal skips of three and four frames. The hypothesis that aggressive videos should have the greatest visual change was not supported.

Class	Mean pixel change, skip=3	Mean pixel change, skip=4
Aggressive	16.32	17.07
Non-aggressive	33.29	35.31

Some non-aggressive videos produced very high pixel variation, including approximately 46.94 and 51.00 in the skip-3 analysis. Pixel change therefore captures motion, camera movement and scene variation rather than aggression-specific semantics.

Pixel Change Analysis — Key Observations
- Higher pixel change does not correspond to aggressive behavior.
- Average pixel change was substantially higher for non-aggressive videos:
- Some non-aggressive videos show very high pixel variation, e.g. 46.94 and 51.00, exceeding most aggressive videos.
- Therefore, raw pixel change captures visual motion/variation rather than aggression-specific information.
- Hence, pixel change alone is insufficient to establish whether temporal information is useful for aggressive-behavior classification.

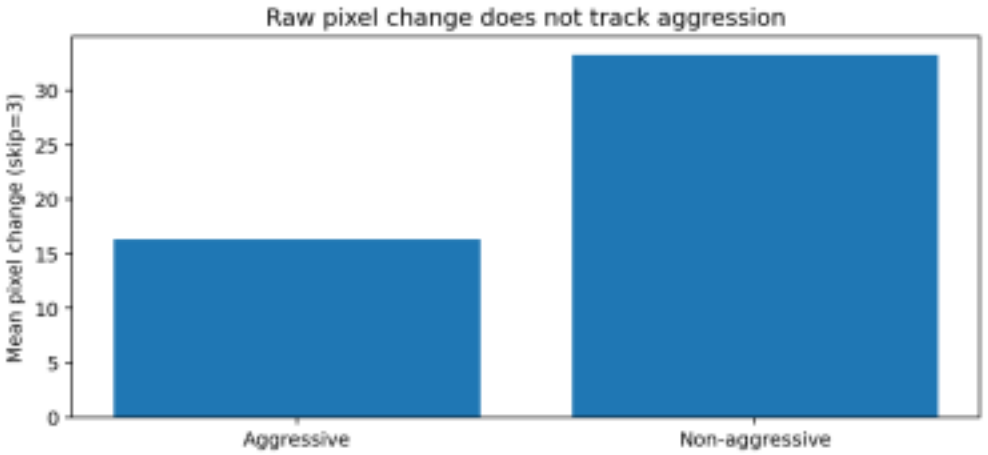


Figure 1. Raw pixel change is not a proxy for aggressive behavior.

3.1 Pretrained visual embeddings

For semantic representation I used SigLIP (google/siglip-base-patch16-224), with CLIP and DINOv2 as comparison encoders. Each frame is mapped to a normalized 768-dimensional representation, preserving frame order for downstream sequence modeling.

Encoder	Agg-Agg	NonAgg-NonAgg	Agg-NonAgg	Agg separation
SigLIP	0.813493	0.676577	0.646336	0.167158
CLIP	0.800262	0.632136	0.587231	0.213031
DINOv2	0.388547	0.29492	0.213011	0.175536

All three encoders show higher within-class than cross-class similarity. CLIP gives the strongest aggressive-class separation in this small comparison, while SigLIP was used for the main temporal experiment.

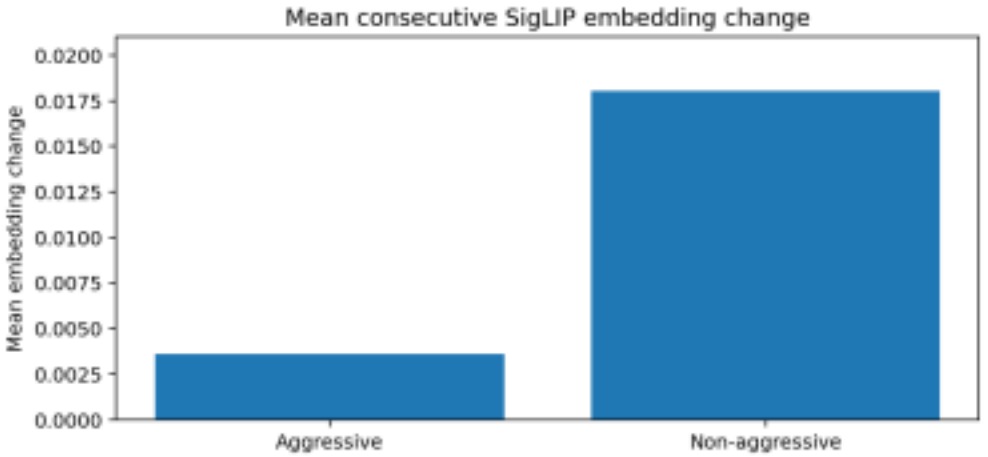
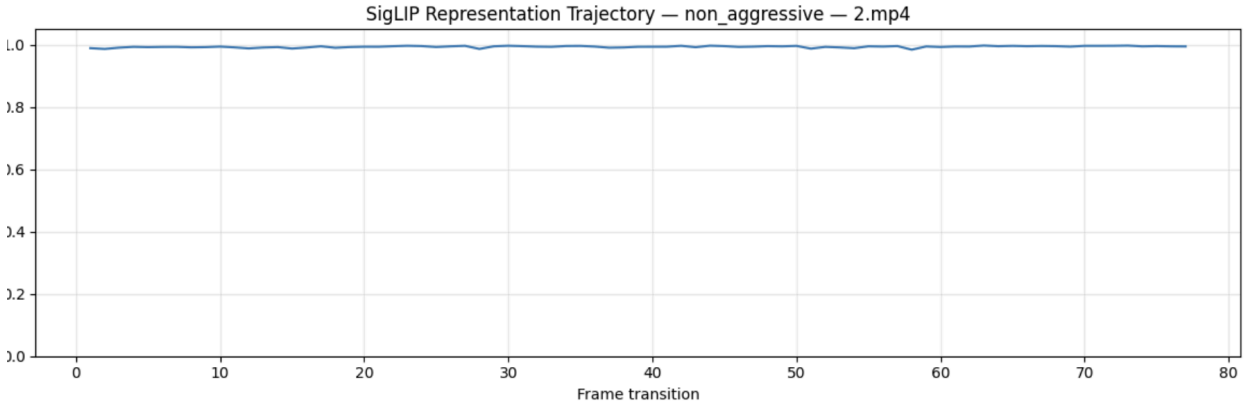
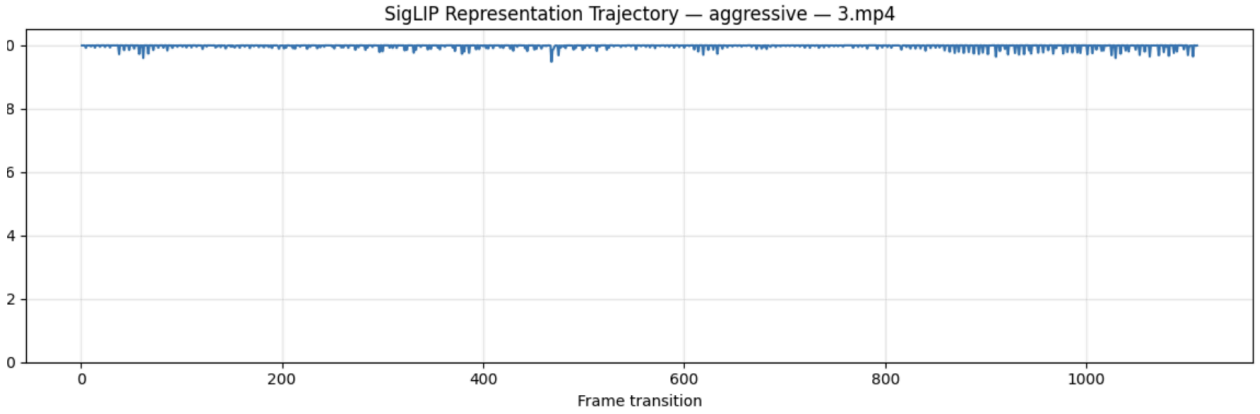


Figure 2. Mean consecutive SigLIP embedding change was higher for non-aggressive videos.

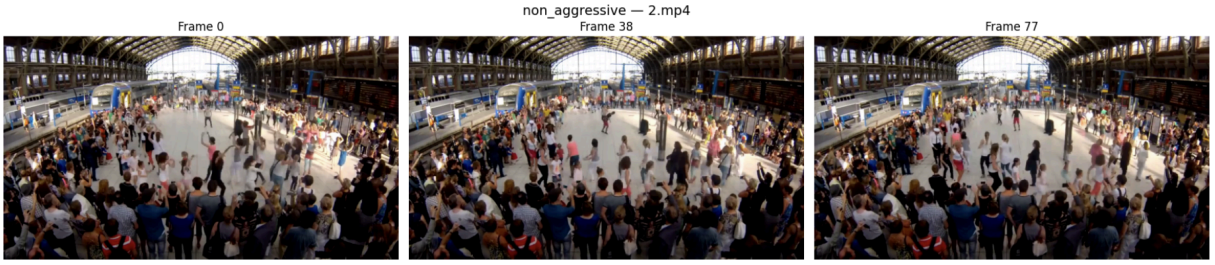
4. Observations:

1.We expected that aggresive videos will have very high variance in embedding change but we are seeing opposite of that instead non-aggressive videos have higher variance in embedding change.
2.But it is evident that there are many disturbances in trajectory of aggressive videos rather than non-aggresive videos.



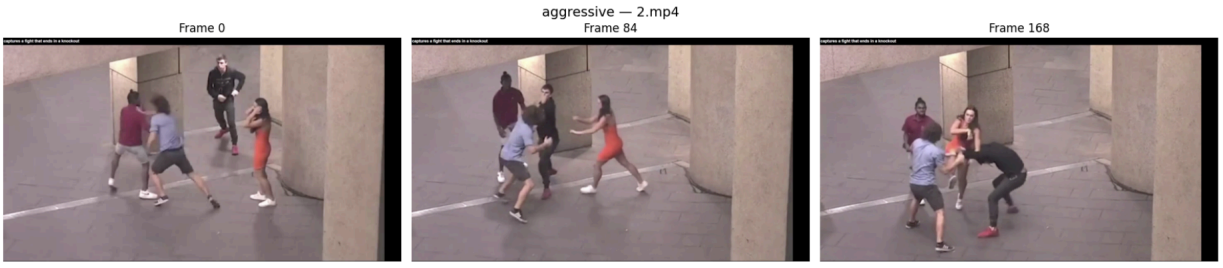



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Class: non_aggressive
Video: 2.mp4
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Class: aggressive
Video: 2.mp4
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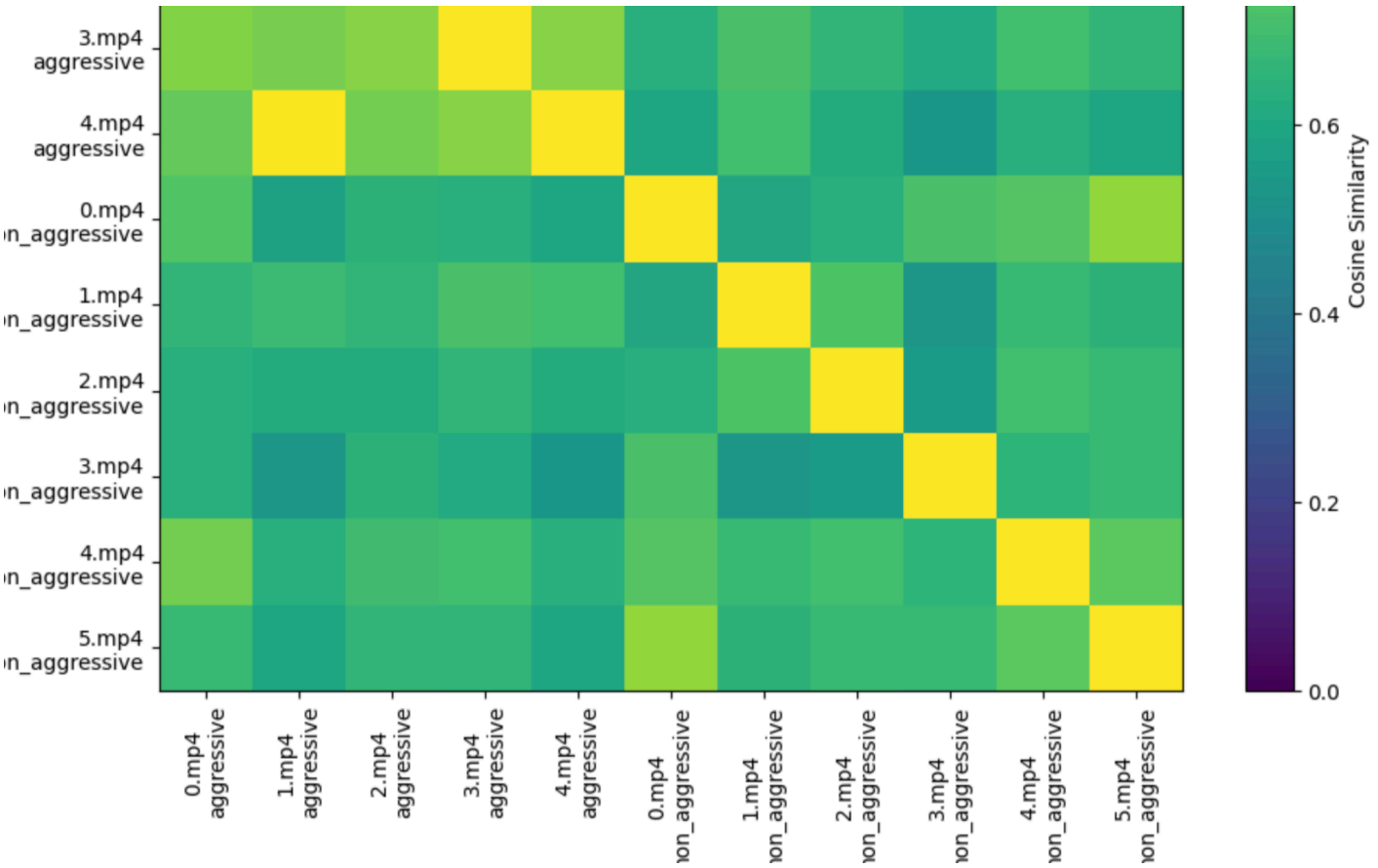



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4. Peaks analysis in trajectory:

Class	Mean num peaks	Mean peak change	Max peak change
Aggressive	115.200000	0.011616	0.035488
Non-aggressive	21.666667	0.025559	0.040531

Video -Level Represenations



Encoder	Aggressive-aggressive	Non-aggressive-non-aggressive	Aggressive-non-aggressive	Aggressive separation	Non-aggressive separation
SigLIP	0.813493	0.676577	0.646336	0.167158	0.030241
CLIP	0.800262	0.632136	0.587231	0.213031	0.044905
DINOv2	0.388547	0.294920	0.213011	0.175536	0.081910

4. Temporal Modeling Approach

To make temporal units comparable across source frame rates, each video was divided into one-second windows with a 0.5-second stride. Sixteen frames were sampled uniformly inside each window. This produced 169 windows: 114 aggressive and 55 non-aggressive.

Split	Aggressive	Non-aggressive	Total
Train	69	20	89
Validation	4	15	19
Test	41	20	61
Total	114	55	169

The split was performed at the source-video level. Training used aggressive videos 0, 1 and 3 plus non-aggressive videos 0, 2, 3 and 5. Validation used aggressive video 2 and non-aggressive video 4. Test used aggressive video 4 and non-aggressive video 1. This avoids frame leakage across splits, but leaves only two source videos in the test set.

4.1 Architecture

Each temporal sample is 16 × 768. A single-layer GRU with hidden size 128 processes the sequence; its final representation feeds a 128→64→2 classifier. Training used weighted cross-entropy and AdamW (learning rate 1e-3, weight decay 1e-4) for up to 30 epochs, restoring the best validation-loss state.

Stage
Video frames + timestamps
1 s windows / 0.5 s stride / 16 sampled frames
SigLIP → 16 × 768 embeddings
GRU, hidden size 128 → temporal representation
2-class classifier → aggressive / non-aggressive

The baseline mean-pools the 16 embeddings into one 768-dimensional vector and trains a small MLP. It therefore receives the same frame information in aggregate but has no explicit mechanism for temporal order.

5. Controlled Temporal Experiments

5.1 Temporal order: ordered vs shuffled

After training, I shuffled the 16 embeddings inside every test window without retraining the GRU. Ten independent random seeds were evaluated. Every shuffle produced the same 100% accuracy, precision, recall, F1 and ROC-AUC.

This is the strongest negative result in the study. The perfect ordered-sequence result cannot be interpreted as evidence of temporal reasoning when destroying the order has no measurable effect.

6. Results

Approach / test	Accuracy	Precision	Recall	F1	ROC-AUC
Temporal GRU — ordered	1.0000	1.0000	1.0000	1.0000	1.0000
MLP	0.9836	0.9762	1.0000	0.9880	1.0000
GRU — shuffled, 10 seeds	1.0000	1.0000	1.0000	1.0000	1.0000

Table 2. Main results. The ordered and shuffled GRU results are reported together because the contrast is central to the conclusion.

6.1 Progressive observation

I evaluated the trained GRU using only the first N observations of each test sequence. One observation gives 0.812 F1, two give 0.988, and four, eight and sixteen all give 1.000.

Observations	Accuracy	F1
1	0.6885	0.8119
2	0.9836	0.9880
4	1.0000	1.0000
8	1.0000	1.0000

16	1.0000	1.0000

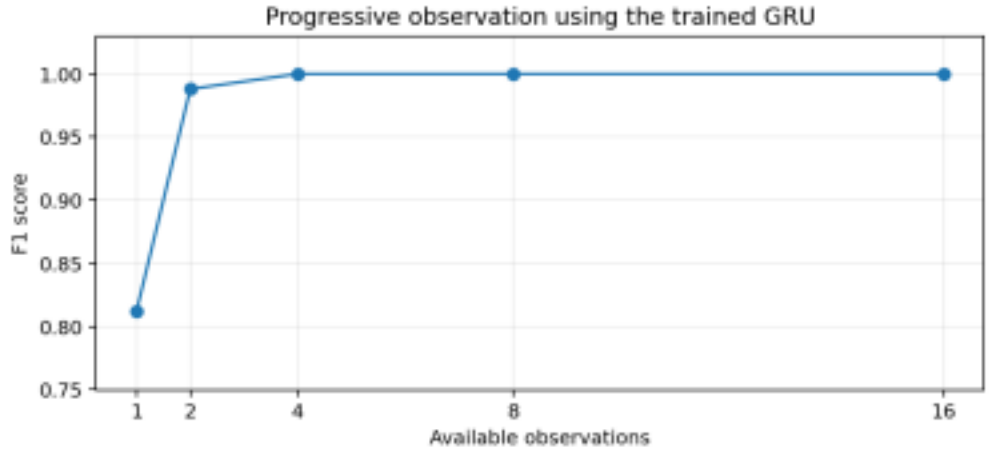


Figure 3. Performance saturates quickly as more observations become available.

This is an observation-availability experiment, not five independently trained models. The model was trained on 16-frame sequences and then evaluated with shorter prefixes. Prefixes also confound observation count with temporal coverage.

6.2 Interpretation

The 98.36% MLP result shows that almost all predictive information is available without order-sensitive modeling. The shuffle experiment then shows that the GRU does not need the order of those observations. Combined with the frame-rate/resolution differences and the two-video test set, this makes static appearance or source-specific shortcuts plausible explanations.

7. VLM Experiment

I used HuggingFaceTB/SmolVLM-256M-Instruct as a small, no-fine-tuning VLM experiment. For each of the same 61 test windows, the original 16-frame temporal window was reduced to eight uniformly distributed chronological frames. All eight were supplied in one multimodal inference call, and the model was asked for exactly one label: AGGRESSIVE or NON-AGGRESSIVE.

The sequence-level VLM remained confused on aggressive clips. I therefore treat this as a qualitative negative result rather than presenting a precise metric that could distract from the main evidence.



In the above case actions of humans can clearly identify that video is aggressive but VLM labels it as non-aggresive



For a non-aggresive videos VLM is able to predict correctly

9. Limitations

• Only 11 source videos; • Aggressive videos are approximately 60 FPS/1920×1080, while non-aggressive videos are approximately 25–30 FPS and lower resolution. • The validation split contains only two source videos, making model selection unstable.
• Progressive observation uses prefixes rather than independently trained models.

• The lightweight VLM was constrained by GPU memory; its result should not be generalized to VLMs broadly.

• The high test score should not be interpreted as evidence of a robust violence detector.

10. What result would convince me that temporal modeling is not useful?

If grouped evaluation on a larger, balanced dataset showed that a strong single-frame baseline consistently matched the temporal model, while reversing or shuffling the sequence had no effect and controlled temporal corruption did not change predictions, I would conclude that temporal modeling adds little value for this task. That would be a positive research conclusion rather than a failed experiment.

11. Future Improvements

The next step should be to improve the evidence before increasing model complexity. A larger dataset with more independent source videos should balance camera characteristics, frame rate, resolution, environment and class. Evaluation should use leave-one-video-out or grouped cross-validation so that performance is measured across genuinely unseen sources. .

• Compare GRU, temporal Transformer and temporal convolution under identical grouped splits.
• Measure prediction change between original, reversed and shuffled sequences.

• Add pose estimation , optical flow or audio as complementary temporal modalities.

• For the VLM branch, use a stronger multi-frame model and ask it first to identify changes between early and late observations.

12. Conclusion

The experiment gives a deliberately cautious answer to the assessment question. More observations can improve the model's decision, but the current evidence does not show that the model needs temporal relationships. The static baseline is almost as strong as the GRU, and destroying temporal order does not change the GRU's predictions. The most defensible conclusion is therefore that this dataset and setup do not yet demonstrate a genuine temporal advantage.

This is still a useful outcome. It identifies exactly what should be tested next: a larger grouped dataset, stronger controls, explicit order-sensitive interventions, and more independent source videos. These changes would make it possible to distinguish genuine temporal reasoning from simply receiving more visual evidence.

Reproducibility note

The work was developed in a GPU-enabled Kaggle notebook. The notebook contains dataset inspection, embedding extraction, time-normalized window construction, GRU/MLP training, shuffled-order evaluation, progressive observation evaluation and the sequence-level VLM experiment. The report follows the assessment's emphasis on experimental design, temporal reasoning, evaluation discipline and research thinking.

Source basis: supplied assessment specification, the submitted Kaggle notebook, and the supplied technical-analysis PDF. No external claims were required for the reported experimental results.